

Module 12: Financial and Operating Leverage

Topic 217: Concept of leverage

The Leverage is actually a term which is taken from the point of view that using a lever, we can enhance our force or our strength meaning that if we use lever and can magnify it by applying even less force.

The concept of **Leverage** basically is that how we can maximize or enhance the profit of our shareholders. In one way, we try to earn more for the shareholders through this enhancement of profit but on the other side, the concept of Leverage is with reference to the Capital Structure. It means that how we can increase or decrease this profit or return by changing the capital structure.



Leverage

- Operating and Financial Leverage
- Use of fixed cost to enhance the return
- Leverage is due to Debt
- Debt generally has fixed cost

Operating and financial Leverage:

So, **Leverage** is actually about the cost; the cost of the debt or the cost of debt financing. This cost of debt financing is actually fixed and when we say that Leverage is creating and actually this Leverage is created through fixed cost.

One fixed cost is our financial fixed cost and in addition to it we have an operational cost which we name Operating Expenses and see it with reference to the cost of goods sold or cost of production.

We have got two types of costs; one is **Financial Cost** and other one is **Operating Cost**. As a result we can have two different Leverages Financial

Leverage and Operating Leverage. The **Operating Leverage** is actually the Leverage that is created because of the fixed operating cost and **Financial Leverage** is the Leverage which is created by because of fixed financial cost or in simple words the fixed rate of interest.

Use of fixed cost to enhance the return:

We can define Leverage as how we can increase the return of shareholder or equity holders by using fixed cost and that is the concept of Leverage. We will consider both the operating profit or profit after tax and return of shareholders or return on equity in it.

Leverage is due to Debt:

If we are focusing on financial Leverage then we say that this **Leverage** is because of Debt meaning that if we have to create financial Leverage then it will need fixed cost and there is not fixed cost of equity but the **fixed cost is always of the Debt**.

So, this means that if we want to create financial Leverage then we have to have some financing which have got some fixed cost and it is only debt that has got fixed cost.

Debt generally has fixed cost:

The Debt having fixed cost is something that creates Financial Leverage. On the other side, fixed operational cost and expenses create Operating Financial Leverage.

So, we say that if a firm wants to create Operating Leverage then it has to fixed one portion of its operating cost and if a firm want to create Financial Leverage then it has to fix the cost of one component of capital which is actually the cost of debt that can be fixed.

We can conclude our basic initial discussion that **Leverage** is actually dependent on the fixed cost. This fixed cost is related to the operations and the Leverage created from the operations will be called **Operating Leverage** and this will affect our profit means it will increase our profit.

And that Leverage which arises from our fixed financial cost which we call **Financial Leverage** and this Financial Leverage increases our equity holders or return on equity.

Module 12: Financial and Operating Leverage

Topic 218: Fixed and Variable Costs

There are **two types of costs** which are almost always there in every business organization.

In this lecture, we will discuss that what **Fixed cost** is and what is the **impact of Fixed cost** and similarly we will also talk about what is **Variable Cost** and what is the **impact of Variable Cost**.

The **Fixed cost** is one which **is fixed** meaning that **cost which never changes** but this is not the right concept about the Fixed cost **as there is no cost in the world which is fixed forever.**

But when we talk about the term Fixed cost which has specific meaning that let's see it.

Fixed Cost

- The cost that does not change with the change in the level of operations.
- Period cost
- Per unit fixed cost changes



Fixed cost:

- The cost that does not change with the change in the level of operations meaning that the cost which does not change with the increase or decrease of production/sales which we name Fixed cost.

The **Fixed cost** can be changed with the influence of some other factors i.e., rent of a building is a fixed cost; if we are doing production in a factory building then the rent of that building is a Fixed cost because there is no relationship between the production and the rent of building and the rent of that building will remain same even we do more or less production in it. So, the rent will not fluctuate with the fluctuation of production therefore rent is called as Fixed cost.

But rent remains same every year or not. No, we have an increase in the rent gradually; generally the contracts are reviewed as every year the amount of rent increases. But that increase is not because of the level of production and this is the point that we need to keep in our mind.

So, the **Fixed cost** is the cost which does not change with the change in the level of operations.

- Fixed cost is the **Period cost** meaning that it is the cost of one month or a year and that cost has to vary with the time but does not vary with the production.

- If any element of the cost is fixed with the time then what will the impact of that; if we pay Rs.100 and produce 100 units with it then our per unit cost will become Rs.1 ($100/100 = 1$) per unit. But if we produce 200 units with this Rs.100 rent then per unit cost or fixed cost will become 0.50 paisa ($100/200 = 0.50$) means become half.

So, we can conclude that the component of Fixed cost will continue to decrease in the per unit cost whenever operations will increase or sales/production will increase

Variable Cost:

Variable Cost

- The cost that changes with the level of operations.
- Marginal cost
- Per unit variable cost does not change



- **Variable cost** is the cost which varies or changes. As we have said that Fixed cost is also changes but we call it fix because it does not change with the level of production or level of operations and if we apply this concept on the Variable Cost then we can say that the cost which fluctuates with the production or operations proportionally and we call this cost the **Variable Cost**.
So, the cost which changes with the level of operations is called **Variable Cost**. For example, if the cost to produce 1 unit is Rs.10 then the cost of producing 2 units will be Rs.20 and likewise if we keep on increasing the number of units, the Variable Cost will keep on increasing.
- Variable Cost is also termed as the **Marginal Cost**. The Marginal Cost means that the cost which we have to incur in order to make further one unit is the Variable Cost but not the Fixed Cost.
- Unlike the Fixed Cost, the **Variable Cost in terms of per unit variable cost does not change** meaning that when we say that the cost of producing 1 unit is Rs.2 then to make 2 units, the cost will be Rs.4 as per unit cost will remain same as Rs.2. **Unlike the Fixed Cost**, as the number of unit increases, the cost in terms of per unit cost decreases.

So, we can conclude our discussion that **Fixed Cost** is a period cost that does not change with the level of production and per unit fixed cost keeps on reducing with the increase in the production that creates the operating leverage.

On the other hand, **Variable Cost** is the cost that proportionally changes with the level of production, it also called the Marginal Cost but the per unit variable cost remains the same.

Module 12: Financial and Operating Leverage

Topic 219: Contribution Margin and its Significance

We are continuing our discussion and we see that when a firm or business increases its production or operations from a certain level then which cost it has to incur further in order to increase these operations or production from that level.

We have talked about that Variable Cost is also called the Marginal Cost that is the cost to produce one more unit which means that if we are operating at a certain level and we want to go above that level then we will have to incur the Marginal Cost and that Marginal Cost is Variable Cost.

So, we can say that the additional cost which has been incurred with the increase of our production is only Variable Cost and there is no change in the Fixed Cost.

It becomes important and relevant for decision making that we should see that what will be the additional profit if we are increasing our production or levels of operations.

Its answer is that when we go in the operations from a given normal level then our Contribution Margin is our Additional Profit. The Contribution Margin is basically the difference of Revenue and Variable Cost meaning that if we subtract our Variable Cost (except Fixed Cost) from our Revenue or Sales then the obtained figure is called the Contribution Cost.

Contribution Margin Formula:

- Difference of Revenue and Variable Cost
- **Formula:**

$$CM = \text{Revenue} - VC$$

Where 'VC' is variable cost and 'CM' is contribution margin.



Contribution Margin Formula = Net Sales - Total Variable Expenses

Significance of CM:

When we go in the operations from a given normal level or do additional production or accept additional orders then for which we have to incur additional cost is only the Variable Cost.

The **Contribution Margin** is actually the name of Additional Profit which is above the normal level. And so, to accept the decision of increasing the production when certain Revenue is offered to us, will be dependent on the Contribution Margin.

It means that we do not compare additional revenue for decision making with our profit but with the Contribution Margin and that is why it becomes significant.

So, Additional Profit in any additional activity is **Contribution Margin**.

CM Ratio:

- The ratio of CM and Revenue/Sales
- Formula:

$$\text{CM Ratio} = \text{CM} / \text{Sales}$$

$$\text{CM Ratio} = 200 / 500$$

$$\text{CM Ratio} = 40\%$$

CM and Variable Cost:

$$\text{VC Ratio} = 1 - \text{CM Ratio}$$

CM Ratio = 1 – VC Ratio

Module 12: Financial and Operating Leverage

Topic 220: Break Even Point

Some people think that Break Even Point is not a very good thing for an organization because it is a point where there is no profit and no loss and every business is meant to be earning profit. It means that if our all businesses are for earning profit and we have a situation where there is neither profit nor loss then what is the benefit of that business.

Break Even Point:

Breakeven

- No profit and no loss
- Self-sustainability
- Contribution is equal to fixed cost



No profit and no loss: Break Even Point is defined in one way that it is a point where the business is neither getting any profit nor any loss.

Self-sustainability: But there is another way of looking at it that Break Even Point tells us about the self-sustainability meaning that if there is any organization which has making no profit and self-sustainability is more important for it which means that it is generating enough revenue therefore they don't need to take funds from outside. So, the positive thing about the Break Even Point is that it tell us about self-sustainability.

Contribution is equal to fixed cost: Another way of talking about the Break Even Point is that this is the point where the contribution or the contribution margin becomes equal to the fixed cost. We can calculate Contribution margin by

subtracting Variable cost from the Revenue (**Contribution cost** = Revenue - Variable cost) and if this Contribution become equal to the fixed cost then **Contribution Margin – Fixed cost = 0** because both are the same and hence that will be the Break Even Point.

So, above these are the three ways of explaining or interpreting the Break Even Point.

Break Even Point Formula:

We have different formulas of calculating Break Even Point. One way to do is to calculate the Break Even Point in terms of rupees or dollars that is in terms of Sales meaning that how much sales should be generated as a result of which our business reaches at such point where its cost and expenses becomes equal so, that is called the Break Even Sales Level.

Break-Even Sales Formula

$$\begin{array}{c} \text{Break-Even} \\ \text{Sales} \end{array} = \frac{\text{Fixed Costs}}{\text{Contribution Margin Percentage}}$$

$$\begin{array}{c} \text{Break-Even} \\ \text{Sales} \end{array} = \frac{\text{Fixed Costs} \times \text{Sales}}{\text{Sales} - \text{Variable Costs}}$$



Break Even Point in units formula:

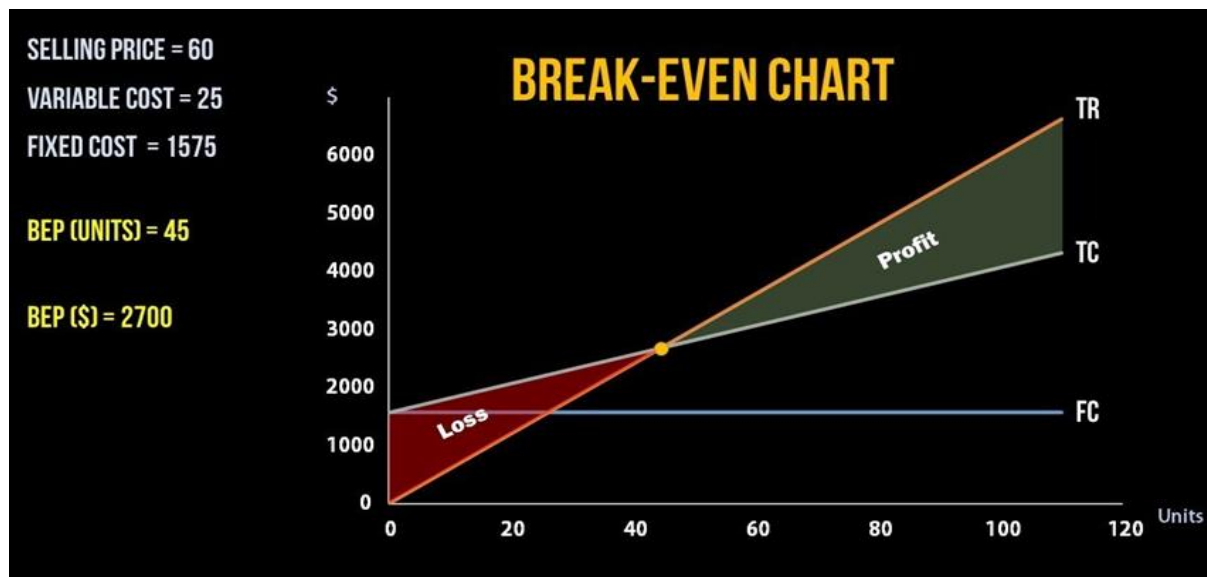
It means that any business needs to sale how much units in order to do Break Even Point meaning that it reaches at no profit and no loss point.



$$\text{Break even point in units} = \frac{\text{Fixed Costs}}{\text{Contribution Margin}}$$



Break Even Point Graph:



Where 'FC' is the fixed cost, 'TR' is the total revenue and 'TC' is the total cost which is the sum of fixed and variable cost.

We can conclude our discussion of Break Even Point that **Break Even Point** is more significant level and it is significant with reference to that this point tells us

that the firm or business has to reach at least in first step where we will avoid losses and here **loss** means that we have to inject some funds in our business in every year otherwise business will not sustain at its level that is a significance of this Break Even Point.

And the second thing as we have discussed that **Break Even Point** is the point where business becomes self-sustainable, it does not require any funds from outside and as when our sales or units sold will go above this point then our profit will start to increase. And the graph also indicates that as our number of units sold or production will increase then we can see in the graph that the difference of our revenue and cost also get increases and the business will have more and more profit.

Module 12: Financial and Operating Leverage

Topic 221: Cost Volume Profit (CVP) Analysis-1

CVP (Cost, Volume and Profit) Analysis:

This topic is related to three very important components which are Cost, the quantity or Volume which refers to the number of units which are produced or sold and Profit which is the prime objective of every business organization.

Cost is the amount incurred to produce:

When we use the term '**Cost**' refers to the *amount that is incurred to produce or manufacture some units*. For production we need to incur Cost and this cost is then further also incurred to sell the units in the market.

Volume is the number of units produced:

The '**Volume**' refers to the number of units which are produced. Why Volume element is considered in this analysis because this is the key component in the discussion of CVP (Cost, Volume and Profit) Analysis.

Higher volume means more production in terms of number of units produced actually reduces the per unit cost of the product. We have discussed earlier that fixed cost element is such which actually reduces in terms of per unit cost when we increase number of production. Because this cost remains cost and when production increases, it is spread over more number of units. So, the component of Volume gets lot of importance because higher or bigger are the volumes, lesser will be the fixed cost per unit and so, lesser will be the total cost per unit.

Profit is the difference of cost and revenue:

We have to see the impact of both on profit because profit is nothing but the difference of revenue and cost. More is the revenue, more will be the profit and less will be the cost, more will be the profit.



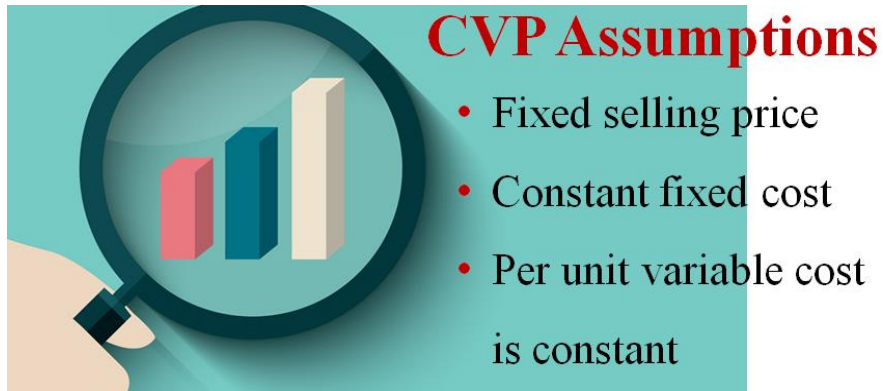
In **CVP (Cost-Volume-Profit) analysis**, we try to see that what are the changes in the variable and fixed costs and how these two things affect the firm's profit when the volumes are changed.

CVP Assumptions:

This CVP analysis has got certain assumptions on the basis of which we say that what will be the impact on the profit if the volume increases and if the volume increases that means the revenue will increase and that means the total cost will also increase. And when the total cost is increasing then there is one component of total cost which does not change.

So, with the change in the volume then we need to remember that one part or portion of the cost changes but the other part does not change and then we have to

see the combined effect of the both Fixed cost and Variable cost. But certainly if we are producing more units and we are selling all those units then the revenue will also increase and we have to see the impact of this on the profit.



Fixed selling price: The first assumption of CVP (Cost-Volume-Profit) is that fixed selling price is there so, per unit selling price does not change. If the price is Rs.10 or Rs.20 per unit then we assume that this remains the same. Only that if the volume goes up then the total of revenue that we will get will be more.

Constant Fixed cost: The second assumption of CVP (Cost-Volume-Profit) is that the total fixed cost remains constant. As we have said that fixed cost is not fixed forever and we say fixed cost because this cost does not change with the change in the level of operations/productions/sales. This is a fair assumption that fixed cost remains constant for a certain period of time when we are performing this CVP (Cost-Volume-Profit) analysis.

Per unit variable cost is constant: the last assumption is that per unit variable cost remains constant and this is a realistic assumption because in short run the per unit variable cost does not change.

Now the question is that what is included in the variable cost then the answer is that raw material or direct material cost is a part of the per unit variable cost, the labor which is directly working is also the part of per unit variable cost. And there is a possibility that over the time these costs go up or down.

So, these above three assumptions are the basic assumptions of the CVP (Cost-Volume-Profit) analysis.

We can conclude our discussion that CVP (Cost-Volume-Profit) analysis is an analysis of these three components Cost, Volume and Profit. We see in this analysis that with the change in volume that means a number of units produces and sold then what is the impact on the revenue and as a result what is the impact on the cost and ultimately what is the impact of this change on the profit.

Module 12: Financial and Operating Leverage

Topic 222: Cost Volume Profit Analysis-2

As we have discussed in our previous lecture that CVP (Cost-Volume-Profit) analysis is basically an analysis that revolves around volume that is quantity; how many units are produced and sold and if units produced and sold increase then what is the impact on the profit and on the cost.

We have also talked about that cost is segregated into two components that is Fixed and Variable cost. And before that we have talked about the impact of fixed and variable cost.

Profit Equation

If we have not segregated or do not separate costs into fixed and variable cost then Profit equation is:

$$\text{Revenue} - \text{Cost} = \text{Profit}$$

And here when we use the term cost then we also mean the operating expense/cost of production.

But here we have segregated the two costs (Fixed and Variable cost) for the purpose of CVP (Cost-Volume-Profit) analysis and we can see that Profit equation is like:

$$\bullet \quad R - \text{TVC} - \text{FC} = \text{P}$$

Where:

R = Revenue

TVC = Total variable cost

FC = Total Fixed Cost

P = Profit

Profit Equation Application

- $(SP \times Q) - (PUVC \times Q) - FC = P$

- $(50Q) - (30Q) - 200,000 = P$

When we take P = 100,000

- $(50 \times Q) - (30 \times Q) - 200,000 = 100,000$

- $(50 \times Q) - (30 \times Q) = 100,000 + 200,000$

- $(50Q) - (30Q) = 300,000$

- $20Q = 300,000$

- $Q = 300,000/20 = 15,000 \text{ units}$

Where:

Q = quantity

SP = per unit selling price

PUVC = per unit variable cost

FC = fixed cost

P = profit

It means that if we are given above all these values then we need to sell 15,000 units which mean that if we want to earn more profit then we can change the value of profit and then see what will be the quantity that will be required.

Example: Breakeven

The same equation to apply it if we want to see that if there is no profit or want to see Breakeven point like how many units need to be sold if we want to see the Breakeven point. It means that in the above same equation if we take Profit as zero (0) as Breakeven point is actually that point where our profit and loss finishes or become zero.

- $(50 \times Q) - (30 \times Q) - 200,000 = 0$
- $(50Q) - (30Q) - 200,000 = 0$
- $(50Q) - (30Q) = 0 + 200,000$
- $20Q = 200,000$
- $Q = 200,000/20 = 10,000$ units

It means that if we want to see with the above given values that how much units we have to produce on which we do Breakeven then the equation help us to determine that.

Module 12: Financial and Operating Leverage

Topic 223: Operating Leverage

We have talked about the concept of Leverage and we have discussed that

Leverage is actually the use of fixed cost which results into the enhancement of return or profit.

And we will taking forward the discussion of Leverage and will see that how many types of Leverage and how it is measured and what changes have happened in the risk due the Leverage.

Leverage Defined:

Leverage is the fixed cost which we have in our system either as the Operating cost or as the Financial Cost and as a result this affects our profit, it enhances our profit or return.

Operating Cost

- The cost incurred to operate or run a business, including:
 - Cost of Sales
 - Sales, General & Administrative-SG&A Expenses
- Does not include **Financial Cost**



That part of the fixed cost which is related to operations meaning that our costs are of the two types; one is **financial cost** and other is **operating cost** which is incurred to make product, to sell and to administer that product and this part of the cost is called the **Operating Cost.**

The Operating cost includes:

Cost of Sales: The cost of goods sold, cost of sale that is the direct cost incurred on the production of the product.

Sales, General & Administrative-SG&A Expenses: The second cost which is part of our operating expenses is called SG&A (Sales, General & Administrative) expenses so, operating cost includes these two items.

But Operating Cost does not include that component which creates the Financial leverage and that is called the financial cost meaning that **Operating Cost does not includes Financial Cost**.

Operating Leverage – OL:

The **Operating cost** is something which creates a Leverage which is called Operating Leverage (OL) because it is operating cost and creates leverage therefore with this reference, we call it Operating Leverage.

So, the basic definition is that the **use of fixed operating cost to enhance profit** is called the **Operating Leverage (OL)**.

How does OL (Operating Leverage) Work?

How does OL Work?

- Fixed cost is the period cost
- When production increases per unit fixed cost decreases
- Higher component of Fixed Cost creates higher OL



When we have a certain component of fixed cost in our operating cost and we increase the production, as a result the fixed cost remains constant and the variable cost increases in proportion to the sales.

When one component of cost remains constant then the increase in the overall cost is not the proportional of the revenue or sales but less than it. So, at higher level of sales or revenue, we find that the profit is even much higher meaning that does not increases proportionally but will increase more than that proportional increase in profit.

The **reason** is that Fixed Cost does not increase with the increase in the level of sales and production.

Fixed cost is the period cost: As we have discussed that if we are paying the rent of building for one month then there will be no effect on that rent of building that we produce how much units even more or less.

When production increases per unit fixed cost decreases: When production increases then this per unit fixed cost keeps on decreasing meaning that if Rs.100 fixed cost is applied to Rs.100 units then this comes out to be Rs.1 per unit but if Rs.100 fixed cost is divided by 200 units then it comes out to be 0.50 paisa or half a rupee.

So, this means that additional units which we produce absorb lesser and lesser fixed cost and as a result the total amount of profit increases.

Higher component of fixed cost created higher OL (operating leverage): The **Operating Leverage** is something that is dependent on the component of fixed cost and the total operating cost.

We can say that if our operating leverage exists then it is because of fixed cost meaning that if we will not have the element of fixed cost in any operation or production or sale, and if we will have some component of fixed cost then there will be some operating leverage and as we keep on increasing the percentage of operating fixed cost component then the operating leverage keeps on increasing.

Module 12: Financial and Operating Leverage

Topic 224: Financial Leverage

We have talked about the concept of Leverage and in detail we have talked about the Operating Leverage. We have discussed that Leverage is basically the presence of fixed cost meaning there will be fixed cost in our structure of business then it will create leverage.

Leverage is increase of profit or leverage through fixed cost. If that fixed cost component is related to the operations or it is the operating fixed cost then the Leverage created is called the **Operating Leverage**.

And if we are looking at the Financial Cost then the leverage created by financial cost is called the **Financial Leverage** that is the use of fixed financial cost to enhance the return is called **Financial Leverage**.

In both leverages (Operating leverage and Financial leverage), the **basic difference** is that:

Operating Leverage relates to our operating profit so, if operating leverage is increasing that means our operating profit is increasing. And **Financial Leverage** is linked with the return on equity that is the return of the shareholder. If financial leverage increases that actually increases the return on equity of the shareholder.

How Financial Leverage does work?

How does it Work?

- Higher proportion of debt creates financial leverage
- When ROI is more than cost of debt ROE goes up



Higher proportion of debt creates financial leverage:

Leverage is determined with reference to the percentage of debt in the overall capital.

Why we say that increase of debt created leverage?

Because we have defined leverage as the use of fixed cost so, first thing that we have to keep in mind that financial leverage increases due to the debt because financial cost of debt is considered to be constant in terms of percentage meaning that if we have taken some financing in the form of debt or bonds then the return or interest that we have to pay such rate on them will remain fixed. This is called the **fixed financial cost** so, financial cost is fixed in terms of the percentage.

When we have a certain amount of debt, we have to pay a certain amount of fixed financial cost and in other word, the fixed amount of interest. But if we increase the amount of our debt, then we will obviously get more fixed financial cost.

How the presence of fixed financial cost increases the return of our shareholders?

For this, **there is an assumption** that this financial cost will always be less than the earning power or return on investment of that business meaning that if business has a capability to earn 20% profit or return then financial cost should be less than 20% then financial leverage will work positively and due to the increase of debt will increase the profit or return of shareholders.

If there is a reverse situation in any business meaning that our earning power becomes less but cost of debt becomes more means our business has a capability to

earn 10% but our financial cost is 15% then in that case, our financial leverage will not act positively but this is its negative impact.

In that case, if we enhance the amount of debt that fixed cost component will actually reduce the return of the equity holders.

When ROI is more than the cost of debt ROE goes up:

It means that if overall return on the investment will be more than the cost of debt then leverage will be created and leverage will increase our return on equity but if our return on investment will be less than our cost of debt then our leverage will decrease our return on equity.

We can conclude our discussion that **Financial leverage** is the use of fixed cost and fixed cost is associated to debt. And the percentage of interest generally that we pay on debt is what makes the fixed financial cost and if we have got better earning power than the cost of debt then financial leverage works positively otherwise it has got a negative element and this is something which creates risk that we will discuss in next lectures.

Module 12: Financial and Operating Leverage

Topic 225: Degree of Operating Leverage

In previous lecture, we have talked about Leverage and then Financial leverage and Operating leverage and now next question is that how much operating leverage is present in the structure of cost (I.e., operating cost) in any firm or business.

We have discussed one point that higher is the proportion of fixed cost, we will have more operating leverage that is correct. But there is a formula or device to measure the level of operating leverage and we can say that this device is called DOL (Degree of Operating Leverage).

As we can measure the temperature through thermometer or we can measure the length from any yard stick similarly, we can measure DOL (Degree of Operating Leverage) is such instrument through which we can measure the

operating leverage of any business or firm so, we call it **DOL (Degree of Operating Leverage)** and there is a formula that we will discuss in this lecture.



How to Measure Operating Leverage?

- Degree of Operating Leverage-DOL
- Higher is the DOL, higher is the risk

One very important thing that we need to keep in mind is that **DOL (Degree of Operating Leverage)** has a direct link with the risk meaning that when our DOL (Degree of Operating Leverage) increases or the value of DOL (Degree of Operating Leverage) is higher in any business organization that indicate that the organization has a higher level of operating leverage and any organization having higher operating leverage will have a higher level of risk.

Why there is higher level of risk?

Because when leverage increases then the enhancement in profit is many folds with the increase of sales or production in fair situation meaning that if sales will increase 10% then profit will increase 16% or 20% but if we reverse this situation then if sales will decrease then this fixed component will reduce our profit proportionally and that is why we say that **DOL (Degree of Operating Leverage)** is higher that means the risk component is higher.

DOL (Degree of Operating Leverage) Formula:

The formula to calculate the value of DOL (Degree of Operating Leverage) is as:




$$\text{Degree of Operating Leverage Formula} = \frac{\% \text{ Change in EBIT}}{\% \text{ Change in Sales}}$$


Where EBIT = Earnings before interest and taxes

DOL (Degree of Operating Leverage) Formula

DOL (In Rupees) = $\frac{\text{EBIT} + \text{Fixed Cost}}{\text{EBIT}}$

$$\text{DOL} = \frac{500,000 + 800,000}{500,000} = 2.6$$

It means that this 2.6 indicates that with 1% change in the sales, our profit will increase by 2.6% meaning if our sales will increase 10% then our profit will increase 26% and if our sales will increase 100% then our profit will increase 260% which indicates that the increase of profit will be the 2.6 times of the increase of the sales not percent but 2.6 times.

DOL (Degree of Operating Leverage) Formula (Units):

DOL (In Units) = $Q / (Q - Q_{BE})$

$$\text{DOL} = 5000 / (5000 - 4000)$$

$$\text{DOL} = 5$$

Where:

Q = Quantity Produced

Q_{BE} = Breakeven Quantity

We can conclude our discussion that DOL (Degree of Operating Leverage) is a measure of operating leverage and higher will be the DOL (Degree of Operating Leverage) then higher will be the operating leverage and also higher will be the risk. But the firm which is growing whose sales are growing then it will be recommended for that firm to keep its DOL (Degree of Operating Leverage) higher because increase of sales create more chances for the enhancement or increase of our profit.

Module 12: Financial and Operating Leverage

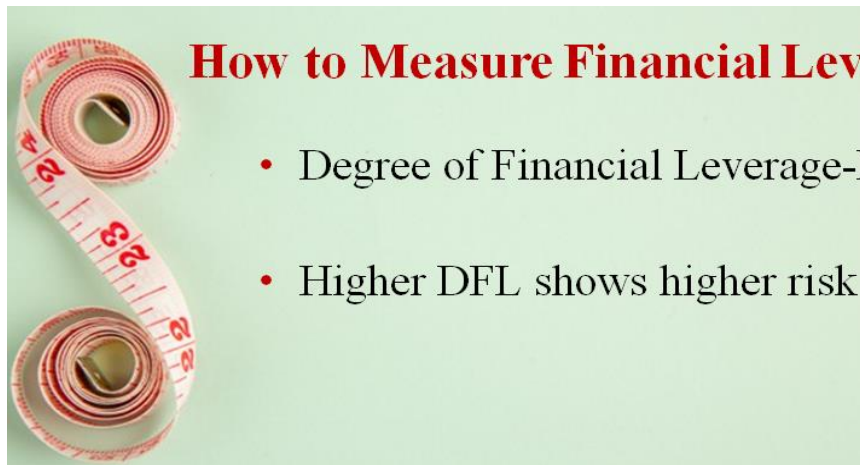
Topic 226: Degree of Financial Leverage

In previous lecture, we have discussed that what leverage is and what are operating and financial leverages and then we talked about DOL (Degree of Operating Leverage) and we saw that how we can calculate the DOL (Degree of Operating Leverage) and what does it show and how it is related to the risk.

Now, let's talk about the other type of leverage which is Financial Leverage. The

Financial Leverage is measured in the same way as DOL (Degree of Operating Leverage), the only thing is that the DFL (Degree of Financial Leverage) has a different formula than DOL (Degree of Operating Leverage) but again we have to see that a higher level of Financial Leverage leads to higher level of risk.

So, we need to consider that if DFL (Degree of Financial Leverage) has to be at a certain point then what will be our risk at that level.



There are two points that we need to keep in mind that DFL (Degree of Financial Leverage) is the measure of financial leverage that is; if DFL (Degree of Financial Leverage) will be more then financial leverage will be more and likewise this DFL (Degree of Financial Leverage) is related to the risk that is; if our DFL (Degree of Financial Leverage) will be more then our risk will also get increase.

The third point is that when it will be more?

The answer is the same that when we will have a bigger proportion of fixed financial cost then we will have higher value of DFL (Degree of Financial Leverage).

DFL (Degree of Financial Leverage) Formula

Degree of Financial Leverage Formula



$$\text{Degree of Financial Leverage} = \frac{\% \text{ Change in EPS}}{\% \text{ Change in EBIT}}$$


$$\text{Degree of Financial Leverage} = \frac{\text{EBIT}}{\text{EBIT} - \text{Interest}}$$


Where **EBIT** = Earnings before interest and tax (it is an absolute measure that it comes in term of rupees or dollars)

EPS = Earnings per share (it is an absolute measure that it comes in term of rupees or dollars)

As **%change in EPS** = EBIT – Interest

So, the relationship is that how much is the %change in EPS with a certain %change in EBIT.

DFL (Degree of Financial Leverage) Formula

$$\text{DFL (In Rupees)} = \text{EBIT} / \text{EBIT} - \text{I} - [\text{PD}/(1-t)]$$

$$\text{DFL} = 500,000 / 500,000 - 200,000 - [100,000/(1-40\%)]$$

$$\text{DFL} = 3.75$$

Where **PD** is preferred share dividend, **I** is the amount of interest on the debt and **t** is income tax rate

We consider PD (Preferred share dividend) in DFL (Degree of Financial Leverage) because the percentage of dividend on the preferred shares is often fixed meaning that fixed percentage of dividend is paid to the shareholders as fixed rate of interest is paid to the bond holders or in the case of debt.

So, if a company has preferred shares and the preferred dividend is fixed in terms of percentage then this formula would look like this.

Why we use (1-t) in this formula?

This is because that the interest is paid from the profit before tax on which there has not been tax charged on it but preferred dividend is paid from profit after tax that means the amount of preferred dividend if it is to be compared with EBIT has to be brought back to the level where the tax has not been deducted out of it.

So, when we divide PD (preferred dividend) by (1-t), we get that amount of preferred dividend which is before income tax.

Now,

$$\text{DFL (In Rupees)} = \text{EBIT} / \text{EBIT} - I - [\text{PD}/(1-t)]$$

$$\text{DFL} = 500,000 / 500,000 - 200,000 - [100,000/(1-40\%)]$$

$$\text{DFL} = 3.75$$

Its interpretation is that with the change of 1% in EBIT, 3.75% change will happen in EPS (Earning per share).

We can conclude our discussion that **DFL (Degree of Financial Leverage)** is a measure of financial leverage and it is something which is related to the risk profile of the company and if we want to increase the degree of financial leverage, we have to increase the amount of debt and so, the cost.

But the last thing which we need to remember that when **DFL (Degree of Financial Leverage)** is increasing, the risk is also increasing because if our fixed cost of debt will be more as compare to our earning power then higher **DFL (Degree of Financial Leverage)** will not go towards the more earning per share but our earning per share continue to reduce with same greater proportion.

Module 12: Financial and Operating Leverage

Topic 227: Degree of Total Leverage

In previous lecture, we have talked about the two leverages that is Operating leverage and Financial leverage and we have also discussed that how do we calculate that how much financial or operating leverage is there in a business organization and we said that we calculate the degree of operating and financial leverage which actually determines the level of operating and financial leverage.

Total Leverage:

Operating leverage is one way of creating leverage and Financial leverage is the other way but obviously in one business organization we do have operating cost which is fixed operating cost and we also have fixed financial cost. So, this means that there must be a combined effect of the two leverages and that is called the

Total Leverage.

What is Total Leverage?

- Change in EPS with reference to change in Sales

Σ
 Sigma

Π
 Pi

If we remember that the operating leverage was actually the link between Sales and EBIT (Earning before interest and tax) and it was the %change in EBIT with the given %change in Sales as:



Degree of Operating Leverage Formula

=



% Change in EBIT

÷



% Change in Sales

And in Financial leverage we related EBIT with EPS (Earning per share) that is:

Degree of Financial Leverage Formula



$$\text{Degree of Financial Leverage} = \frac{\% \text{ Change in EPS}}{\% \text{ Change in EBIT}}$$


$$\text{Degree of Financial Leverage} = \frac{\text{EBIT}}{\text{EBIT} - \text{Interest}}$$


If we see these formulas as combined then Sales to EBIT and EBIT to EPS are two components where Sales to EBIT is actually Operating Leverage and EBIT to EPS is actually the Financial Leverage.

So, when we combine the two that is Operating and Financial Leverages then we get the **Total Leverage**.

How do we combine the two (Operating and financial leverage)?

There are two ways of combining it that either we can add or we can multiply. It means that we can take their product and also take their sum. The answer is that we take the product of two.

Here the product means that we multiply DFL (Degree of Financial Leverage) and DOL (Degree of Operating Leverage) in order to get the **Total Leverage**.

How to measure Total Leverage?

How to Measure Total Leverage?

- Combined effect of Operating and Financial Leverage
- Product of DOL and DFL

?

Total Leverage is the combined effect of Operating and Financial Leverage.

And it is calculated by multiplying the two values of leverages that is DFL (Degree of Financial Leverage) and DOL (Degree of Operating Leverage). It means that

value of DFL (Degree of Financial Leverage) is 3 and the value of DOL (Degree of Operating Leverage) is 2 then our **Total Leverage** will be $2 \times 3 = 6$

As we have said that these two leverages starts from Sales and then EBIT and then to EPS then in Total Leverage, we see the impact of Sales directly on EPS (Earning per share).

DTL (Degree of Total Leverage) Formula

The theoretical part of this formula is that what is the %age change in EPS with a certain %age change in Sales; if Sales increase or decrease 1% then how much percent our EPS increases or decreases and this is called **DTL (Degree of Total Leverage)**; the impact of fluctuation in Sales on the EPS.

DTL (Degree of Total Leverage) = (%age Change in EPS)/(%age Change in Sales)

$$\text{DTL} = \text{DOL} \times \text{DFL}$$

$$\text{DTL} = 2.6 \times 3.75 = 9.75$$

This 9.75 means that with 1% change in sales, our EPS will change by 9.75% and as this value will get increases then this business's Total Leverage will get increase.

$$\text{DTL (In Rupees)} = (\text{EBIT} + \text{FC}) / \text{EBIT} - \text{I} - [\text{PD}/(1-t)]$$

$$\text{DTL} = 500,000 + 800,000 / 500,000 - 200,000 - [100,000/(1-40\%)]$$

$$\text{DTL} = 9.75$$

Where EBIT = Earning before interest and tax

FC = Fixed Cost

PD = Preferred share

t = income tax rate

I = amount of interest

We can conclude our discussion that **DTL (Degree of Total Leverage)** tells us the total effect of leverage in our business or firm and this is a combined effect of both Operating Leverage and Financial Leverage but this combination of DOL (Degree of Operating Leverage) and DFL (Degrees of Financial Leverage) is not in the form of sum but it is the product of DOL (Degree of Operating Leverage) and DFL (Degrees of Financial Leverage).

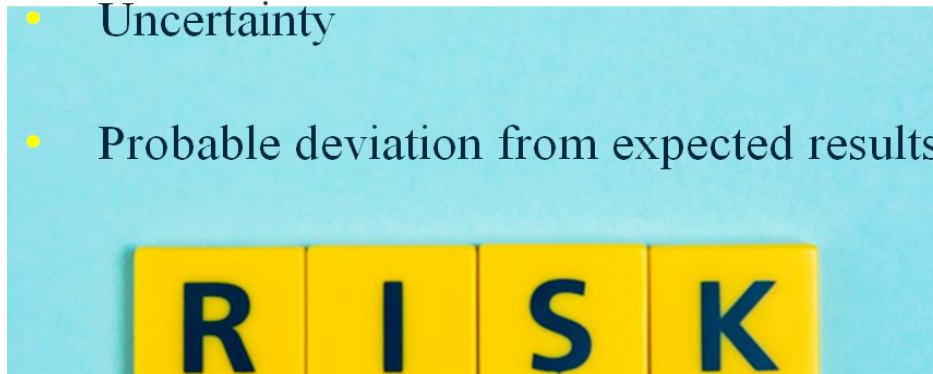
Module 12: Financial and Operating Leverage

Topic 228: Risk and Leverage

We have talked a bit about the relationship of Leverage and Risk and its principal we have said that Leverage increases risk; higher is the leverage whether it is Operating Leverage or Financial Leverage, it actually increases risk.

Risk:

- Uncertainty
- Probable deviation from expected results



One way of defining Risk is **Uncertainty**; the level of Uncertainty that we do not know what is going to happen, how much will be our profit or how much will be our loss, is something that is Risk.

In Risk, it is not just the negative element but there is also a positive element meaning that it is not the name of any loss but there is a possibility of both profit and loss in Risk.

We can also define Risk as the **Probable deviation** or such possible deviation that is from our expected results meaning that we were expecting something or amount of profit or any percentage of profit or any earning per share but how much this earning per share or profit can go up or down from our expectations that is called Risk.

Relationship of Risk and Leverage:



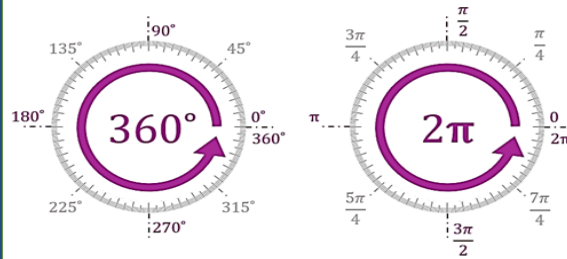
In simple words, leverage enhances risk that means if we increase Leverage then the Risk would automatically go up and if we want to control or curtail risk then we need to reduce our leverage. It means that we have to reduce the component of fixed cost if we want to reduce our risk. And if we are a Risk taker then we would increase the component of fixed cost.

Another important thing that when we use the term Return and as risk is not just a negative thing similarly Return is not always a positive amount that we will receive. The term Return is used on what we get on our investment which can be positive or it may also be negative.

Degree of Leverage Measure Risk:

Degree of Leverage Measures Risk

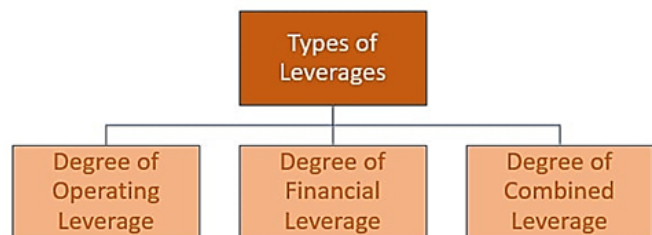
- High DOL, DFL, and DTL indicate higher risk



We have talked about the measurement method of leverages and these three methods are DOL (Degree of Operating Leverage), DFL (Degree of Financial Leverage) and DTL (Degree of Total Leverage) and one principal is true for all these three leverages that is; whenever these leverages DOL (Degree of Operating Leverage), DFL (Degree of Financial Leverage) and DTL (Degree of Total Leverage) will be higher then the Risk will be higher.

Lets revise that we have discussed and relate it to the risk component.

- DOL: Small change in sales can cause big change in Profit
- DFL: Small change in profit can cause big change in EPS
- DTL: Small change in sales can cause big change in EPS



DOL is a small change in sales that causes a big change in our profit. So, higher is the DOL, higher is the risk and as with the increase of sales will increase our profit and with the decrease of sales will decrease our profit.

DFL is a small change in profit which is taken as EBIT (Earning before interest and tax) and a large big change in EPS (Earning per share) meaning that if we enhance our profit, we will enhance a lot our EPS but our profit goes down then our EPS will considerably go down because of DFL.

DTL explains the same thing that a little fluctuation in Sales would lead to a lot of fluctuation in EPS meaning that if Sales are getting to increase then we have an upward trend that's wonderful, DTL should be higher and that is way we make extra profit.

But if the certainty is not there, if the things are volatile as sometimes market goes up or sometimes goes down and if the business is not going up with a smooth trend and there are unknown possibilities of decline or boom then higher level of DTL would be very dangerous.

Because at times where there will be decline in sales that would directly affect EPS (Earning per share).

We can conclude our discussion that Risk is an inherent of such a business organization where there is fixed cost component and it is upto the management of business organization that how much risk they are ready to take.

This should also be considered by considering the overall business environment, economic environment and if there is a lot of stability in the business and economic environment of a country then it is better to have higher degrees of leverages but if we do not have a very stable and growing business or economy then it is recommended we have try to curtail or reduce or keep our leverages at a certain level so, that at times of recessions we may not suffer a lot.